Effects of Extreme Weather Events on Bee Diversity and Community Composition

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Introduction

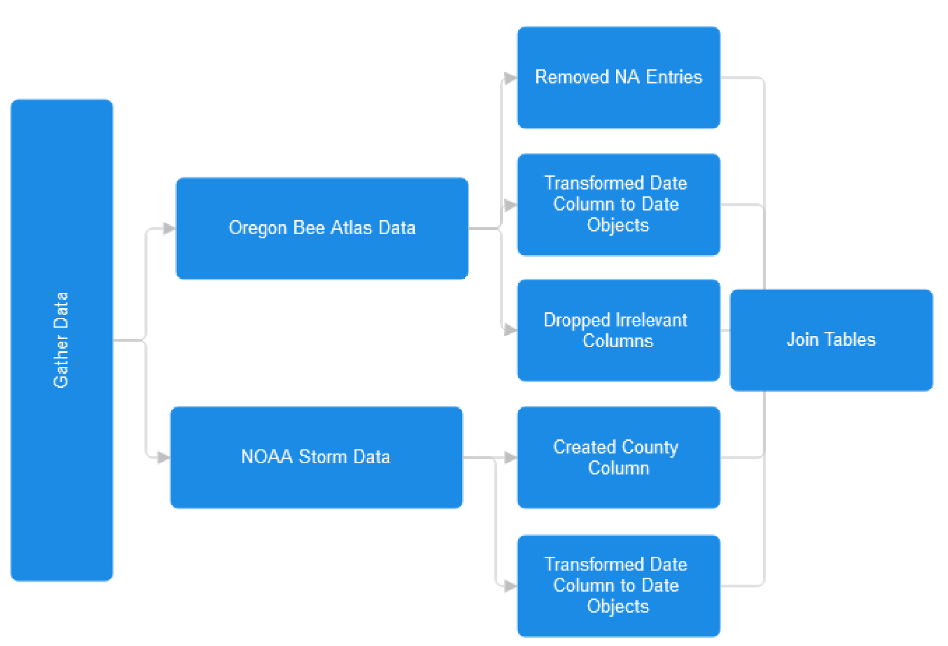
- Extreme weather events are increasing in Oregon (IPCC 2021; OCCRI 2023), and temperature shifts strongly affect bees (Heinrich 1979; Woods et al. 2019).
- Most research focuses on long-term warming (Potts et al. 2016), **not short-term extremes.**
- Specialists may be more vulnerable than generalists (Fowler & Droege 2020; Wilson & Carril 2015).
- **Gap: No analysis on how short-term extremes affect bee diversity and community composition**

Research Questions

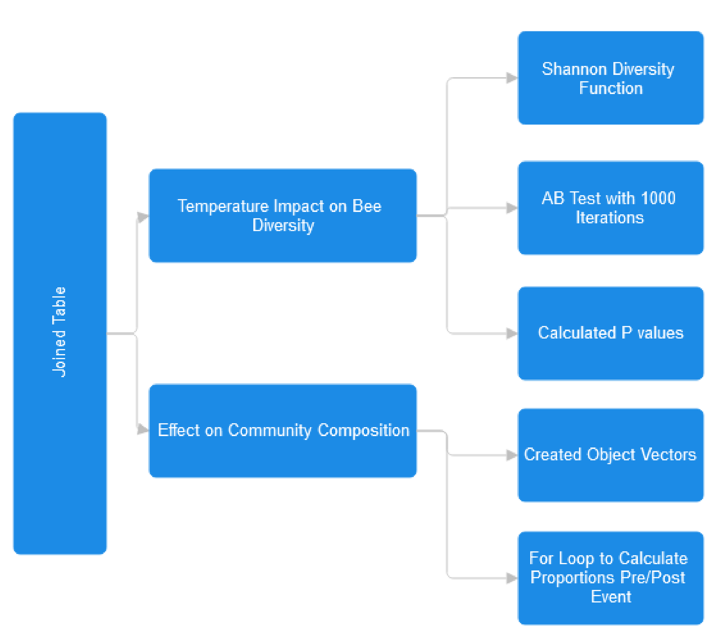
- How does extreme weather events such as extreme heat, cold, and winter storms effect bee diversity?
- How does extreme weather events effect community composition of bees?

Methods

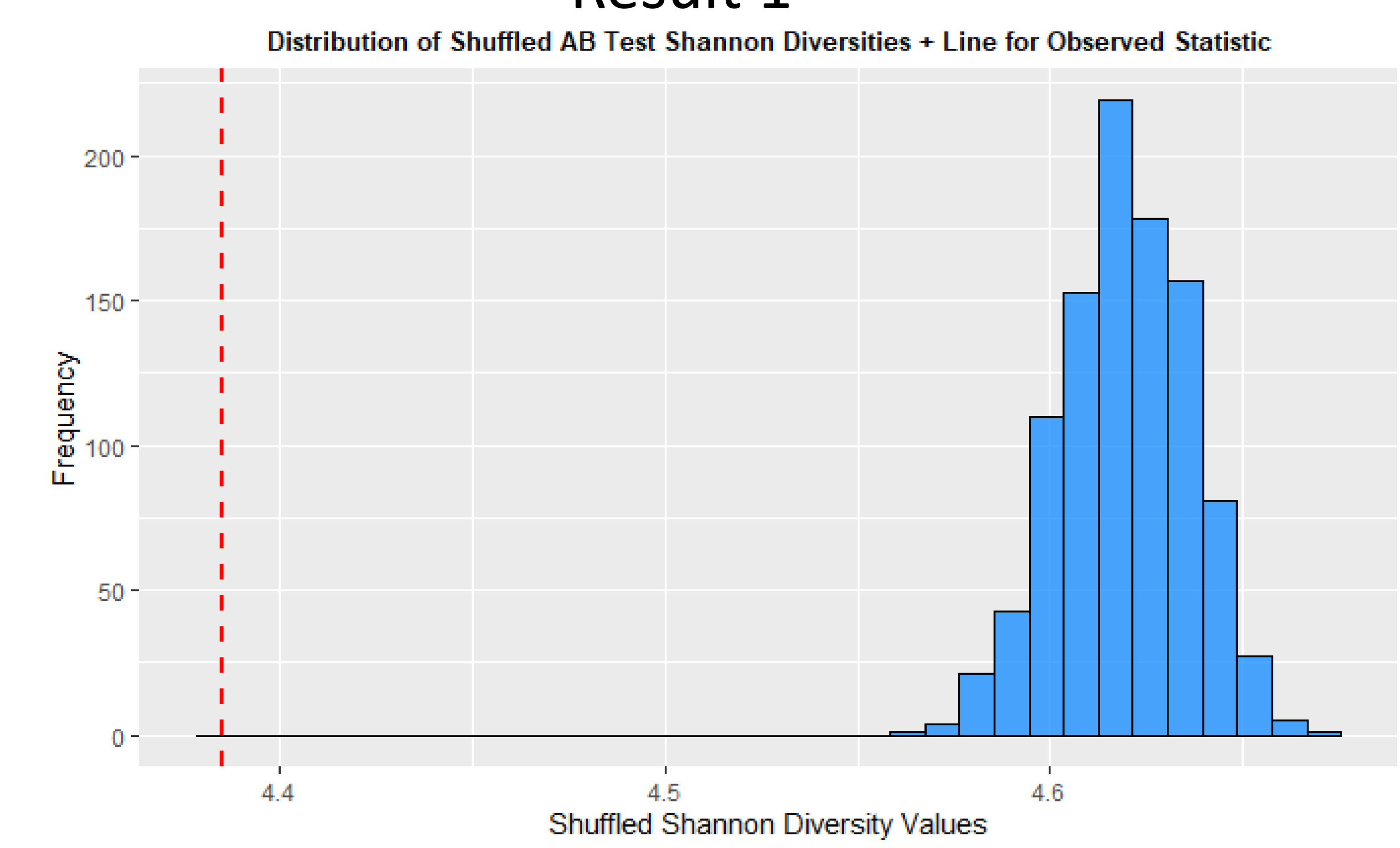
Data Exploration



Data Analysis

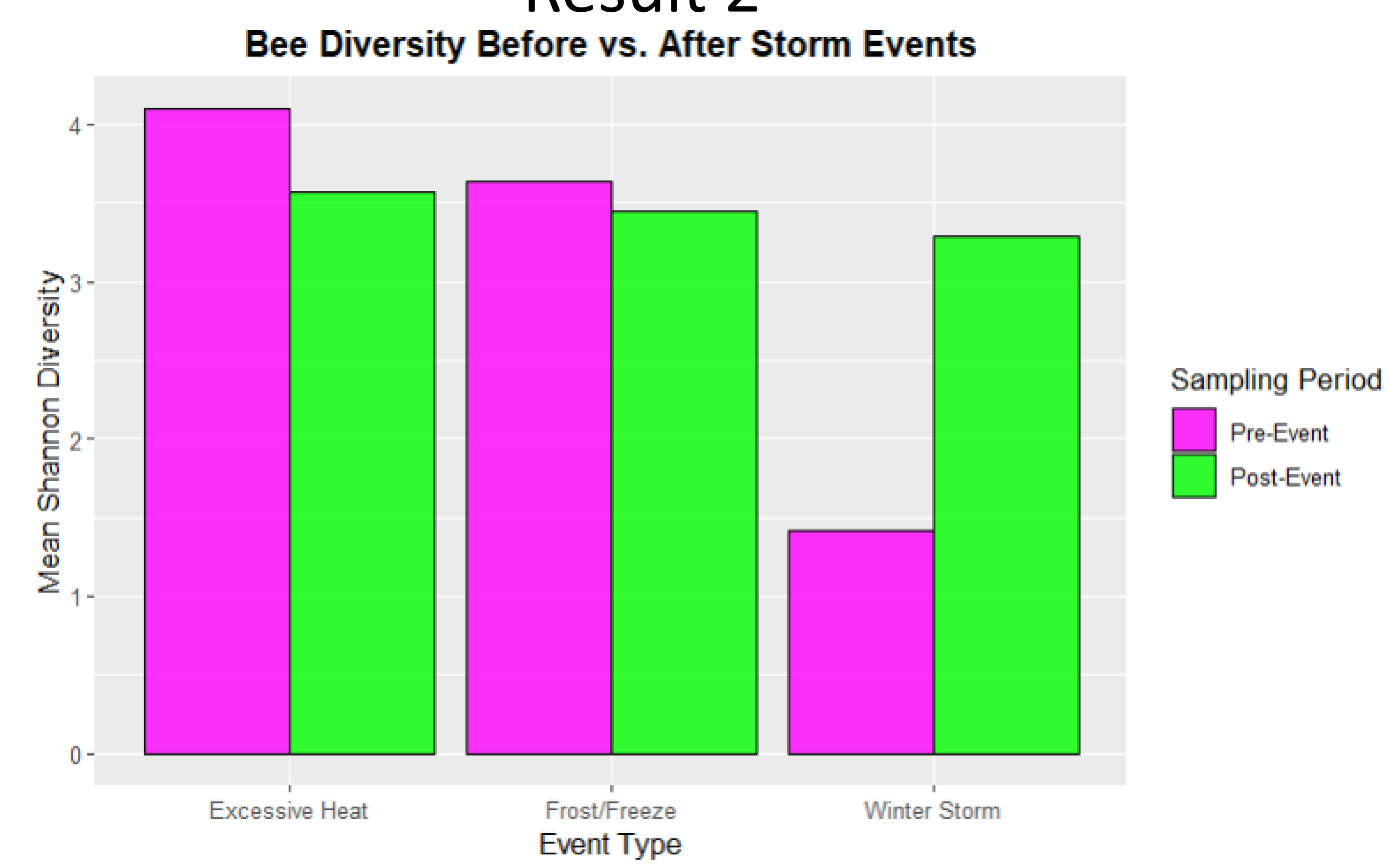


Result 1

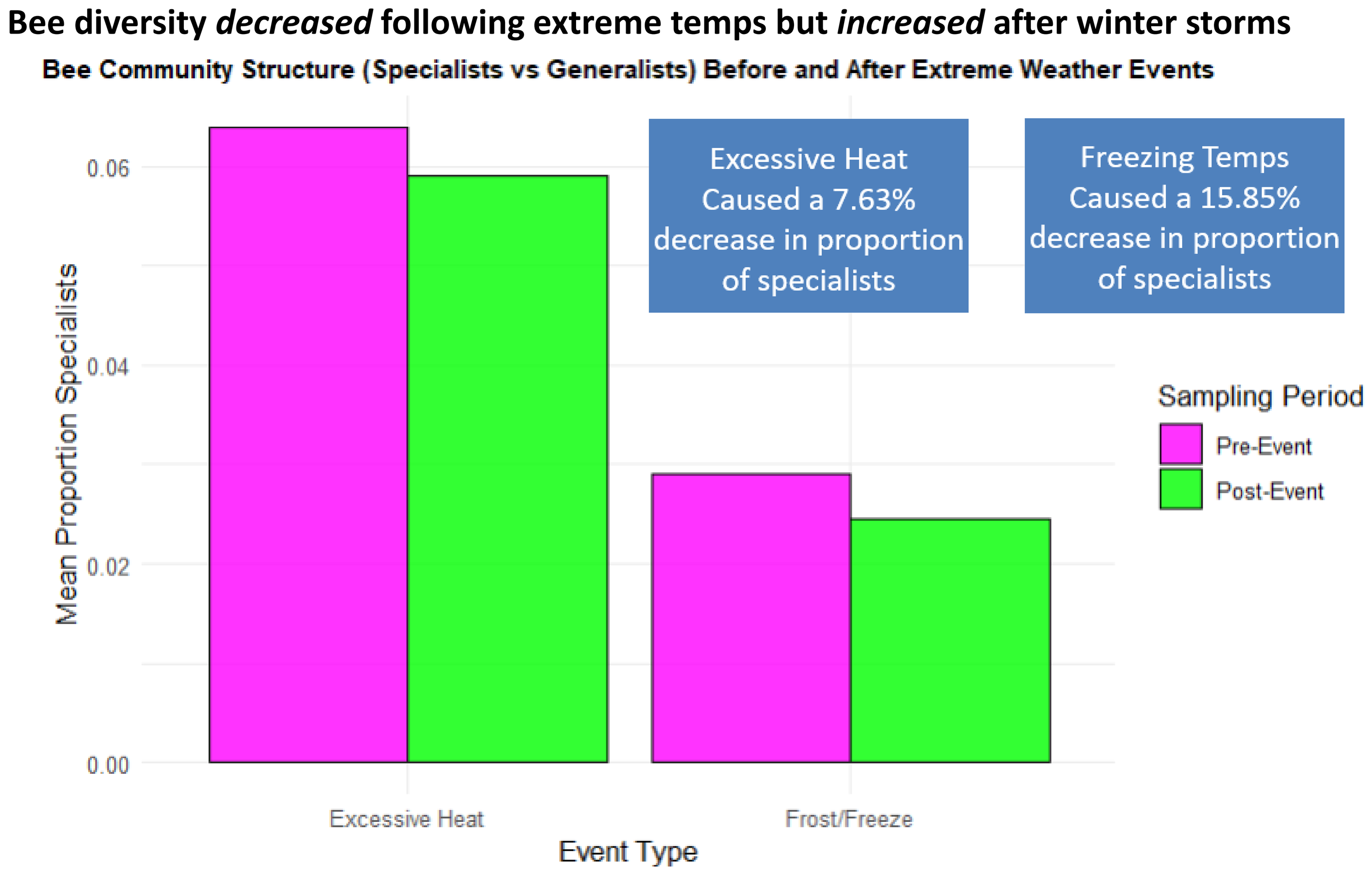


Shuffled AB Test revealed a P-value of 0

Result 2



Bee diversity *decreased* following extreme temps but *increased* after winter storms



Bee community composition shows a clear shift toward generalist species following extreme weather events

Conclusions

- Our shuffled AB test on Shannon diversity showed a highly significant effect of extreme weather (p = 0), demonstrating that bee diversity is not random relative to weather events.
- Extreme heat and cold led to reduced Shannon diversity, while diversity increased following winter storms.
- Analysis of community composition showed that these stress events shifted bee communities toward generalist species, suggesting that specialists are disproportionately impacted by thermal extremes.
- These findings indicate that short-term climate shocks can both reduce diversity and restructure bee communities, with generalists acting as climate-resilient taxa while specialists remain more vulnerable.

References

Danforth, B. N., Minckley, R. L., Neff, J. L., & Fawcett, F. (2019). The solitary bees: Biology, evolution, conservation. Princeton University Press.
Fowler, J. (2016). Specialist Bees of the Northeast: Host Plants and Habitat Conservation. Northeastern Naturalist, 23(2), 305–320. https://doi.org/10.1656/045.023.0210
Franzén, M., & Öckinger, E. (2012). Climate-driven changes in pollinator communities: The return of the dark side of cold winters. Journal of Insect Conservation, 16(2), 215–222. https://doi.org/10.1007/s10841-011-9410-y
Goulson, D. (2010). Bumblebees: Behaviour, ecology, and conservation. Oxford University Press.
Goulson, D., Nicholls, E., Botías, C., & Rotheray, E. L. (2015). Bee declines driven by combined stress from parasites, pesticides, and lack of flowers. Science, 347(6229), 1255957. https://doi.org/10.1126/science.1255957
Heinrich, B. (1979). Bumblebee economics. Harvard University Press.
Intergovernmental Panel on Climate Change. (IPCC) (2021). Climate change 2021: The physical science basis (V. Masson-Delmotte, P. Zhai, A. Pirani, et al., Eds.). Cambridge University Press. https://doi.org/10.1017/9781009157896
Michener, C. D. (2000). The bees of the world. Johns Hopkins University Press.
Mishra, M., Bhunia, P., Shubham, S., Sen, R., Bhunia, R., Gulia, J., Mondal, T., Zidane, Z., Saha, T., Chacko, A., Raj, R., Vyshak, S. S., & Kaushal, L. (2023). The impact of weather change on honey bee populations and disease. Journal of Advanced Zoology, 44(S7), 180–190. https://doi.org/10.17762/jaz.v44iS7.2755
Oregon Climate Change Research Institute. (2023). The Fifth Oregon Climate Assessment. Oregon State University, Corvallis, OR. https://www.climatechange.oregonstate.edu/oregon-climate-assessments
Pereboom, J. J., & Biesmeijer, J. C. (2003). Thermal constraints for stingless bee foragers: The importance of body size and coloration. Oecologia, 137(1), 42–50. https://doi.org/10.1007/s00442-003-1333-6
Potts, S. G., Imperatriz-Fonseca, V., Nguyen, H. T., Bartuska, A., Medellín, R. A., Baste, I. A., Oteng-Yeboah, A., Watson, R., et al. (Eds.). (2016). The assessment report of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services on pollinators, pollination and food production. Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services.
Praz, C. J., Müller, A., & Dorn, S. (2018). Host-plant specialization in solitary bees: A case for polylecty evolution. Biological Reviews, 93(4), 1693–1716. https://doi.org/10.1111/brv.12414
Russo, L. (2016). Positive and negative impacts of non-native bees on native ecosystems. Annual Review of Entomology, 61, 13–31. https://doi.org/10.1146/annurev-ento-010715-023719
Thornton, P. K., Whitfield, S., & Challinor, A. J. (2023). Climate extremes and their implications for global food systems. Annual Review of Environment and Resources, 48, 203–229. https://doi.org/10.1146/annurev-environ-120120-021013
Wilson, J. S., & Carril, O. M. (2015). The bees in your backyard: A guide to North America's bees. Princeton University Press.
Woods, H. A., Dillon, M. E., & Pincebourde, S. (2019). The roles of microclimatic diversity and of behavior in mediating the responses of ectotherms to climate change. Journal of Thermal Biology, 84, 80–92. https://doi.org/10.1016/j.jtherbio.2019.06.0031